

```
1 function [ p, idxs] = paretoFront( p )
2 % Filters a set of points P according to Pareto dominance, i.e., points
3 % that are dominated (both weakly and strongly) are filtered.
4 %
5 % Inputs:
6 % - P      : N-by-D matrix, where N is the number of points and D is the
7 %            number of elements (objectives) of each point.
8 %
9 % Outputs:
10 % - P      : Pareto-filtered P
11 % - idxs   : indices of the non-dominated solutions
12 %
13 % Example:
14 % p = [1 1 1; 2 0 1; 2 -1 1; 1, 1, 0];
15 % [f, idxs] = paretoFront(p)
16 %      f = [1 1 1; 2 0 1]
17 %      idxs = [1; 2]
18
19 [i, dim] = size(p);
20 idxs = [1 : i]';
21 while i >= 1
22     old_size = size(p,1);
23     indices = sum( bsxfun( @ge, p(i,:), p ), 2 ) == dim;
24     indices(i) = false;
25     p(indices,:) = [];
26     idxs(indices) = [];
27     i = i - 1 - (old_size - size(p,1)) + sum(indices(i:end));
28 end
29
30 end
```